

Instructions for Updating ShareGo Final Report

This document provides chapter-by-chapter instructions on what needs to be updated in the FYP-I report ("ShareGo signed final report.pdf", December 2, 2025, 53 pages) to reflect the current state of the ShareGo project as of March 2026. All FYP-II work has been completed.

Reference Files Used for This Analysis

- ShareGo signed final report.pdf (53 pages, FYP-I submission)
- ShareGo_FYP_Project_Documentation.md (project documentation)
- server/app/models.py (18+ database models)
- server/app/api/routes/ (15+ API route modules)
- server/app/core/config.py (server configuration)
- server/app/domain/ (business logic services)
- server/templates/ (11 Jinja2 admin templates)
- server/alembic/versions/ (10 database migrations)
- mobile/lib/features/ (all mobile feature modules)
- mobile/lib/main.dart (Flutter entry point)
- mobile/pubspec.yaml (Flutter dependencies)
- mobile/.env and server/.env (environment configuration)

Chapter 1: Introduction (Pages 1-2)

Priority: Low

Section	Current Content	What to Update
1.1 Purpose of this Document	Mentions "design, development and evaluation"	Update to: "This document presents the complete design, development, implementation, and testing of ShareGo." Add that all backend functionality, admin dashboard, and database connectivity are now fully implemented.
1.2 Intended Audience	Generic description	Optionally add "system administrators" since admin panel is now fully functional.
1.3 Definitions	Lists FYP, AI, UI, UX, API,	Add: JWT (JSON Web

	SDG	Token), OTP (One-Time Password), KYC (Know Your Customer), CNIC (Computerized National Identity Card), CORS (Cross-Origin Resource Sharing), REST (Representational State Transfer), ORM (Object-Relational Mapping), CRUD (Create Read Update Delete)
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Chapter 2: Project Vision (Pages 3-8)

Priority: Medium

Section	Current Content	What to Update
2.4 Goals and Objectives	Lists 7 high-level objectives	Add achieved objectives: Password-based auth, Google Sign-In, multi-LLM AI assistant (Gemini + NVIDIA MiniMax + OpenAI), flight tracking (AviationStack), dark theme, 120Hz refresh rate, audit logging, feature flags, wallet ledger.
2.5 Project Scope	Describes traveler listing, buyer requesting, AI chatbot	Expand to include: Admin dashboard with KYC approval, escrow management, marketplace moderation. Wallet ledger. Booking state machine with OTP-based handover. Dispute management. In-app messaging.
2.9 Stakeholders	Lists Travelers, Buyers, Admin	Expand Admin role: "Admin/Ops with full Jinja2 web dashboard for KYC, booking management, escrow, marketplace moderation, audit logs, and feature flags. Cookie-based session auth."

Chapter 3: Literature Review (Pages 9-15)

Priority: None - No changes required

Chapter 4: Software Requirement Specifications (Pages 16-30)

Priority: HIGH

4.1 List of Features

Add these implemented features to the feature list:

- Password-based registration and login (bcrypt hashing)
- Google Sign-In (OAuth 2.0 ID token verification)
- OTP-based email verification (bcrypt-hashed, single-use, 300s TTL)
- KYC identity verification (CNIC + Passport support for foreign nationals)
- Booking state machine: REQUESTED -> HOLD_PLACED -> ACCEPTED -> PICKUP_OK -> DELIVERY_OK -> RELEASED
- OTP-based handover verification (pickup and delivery) with GPS logging
- Simulated escrow with wallet ledger system
- Multi-LLM AI assistant (Gemini primary, NVIDIA MiniMax, OpenAI fallback)
- AviationStack flight tracking integration
- Dark/light theme toggle with Material Design 3
- 120Hz high refresh rate support (flutter_displaymode)
- Feature flags system for gradual rollouts
- Comprehensive audit logging (before/after snapshots)
- In-app messaging with booking/listing context
- Review and rating system (1-5 stars)
- Dispute management with evidence upload
- Admin web dashboard (Jinja2 templates, cookie auth)

4.2 Functional Requirements - Add to Traveler

- Trip posting with flight number, airline, capacity, and fee
- Trip admin review process (pending_review -> approved/rejected)
- Accept/decline booking requests
- OTP-based pickup verification with GPS logging
- Receipt/proof upload for purchased items
- View wallet balance and transaction history

4.2 Functional Requirements - Add to Buyer

- Item request with specs, target price, weight, time window
- Browse matched travelers based on route and date
- Fund simulated escrow at booking creation

- OTP-based delivery verification
- Cancel booking before traveler accepts
- Open disputes with evidence upload

4.2 Functional Requirements - Add to Admin

- Cookie-based web login with session management
- KYC approval/rejection with reason notes
- Booking timeline and detail views
- Escrow release/refund actions
- Marketplace listing moderation
- Audit log viewing (before/after snapshots)
- Feature flags management
- User and wallet management dashboards

4.4 Non-Functional Requirements - Update Security

- JWT authentication (HS256, 60-minute expiry)
- OTP hashing with bcrypt (12 rounds, single-use)
- Rate limiting via SlowAPI (3/min for OTP, 5/min for login, 10/min for Google auth)
- CORS middleware with configurable allowed origins
- TrustedHostMiddleware for host header validation
- Admin cookie auth with configurable session timeout
- Owner/role-based access guards on all endpoints
- Audit logging for all admin mutations and security-sensitive actions
- Request ID tracking across all HTTP requests
- Waybill format validation (SG-XXXX-XXXXXX)

4.6 Use Cases - Add New Use Cases

Add these use case tables (same format as existing Tables 4.1-4.13):

- UC14: Google Sign-In Authentication
- UC15: Forgot Password (OTP re-issue)
- UC16: KYC Submission (CNIC/Passport + Selfie)
- UC17: Booking Creation with Escrow Hold
- UC18: Pickup Verification (OTP + GPS + Photos)
- UC19: Delivery Verification (OTP + GPS + Seal Match)
- UC20: AI Chatbot Interaction
- UC21: Marketplace Offer Bargaining
- UC22: Meetup Creation and OTP Confirmation
- UC23: Dispute Filing with Evidence Upload
- UC24: Admin Escrow Release/Refund
- UC25: Feature Flag Toggle

4.9 Database Design - MAJOR UPDATE

ER Diagram (Figure 4.6) and Data Dictionary need major updates.

Update existing tables:

Table	New Fields to Add
User (Table 4.14)	password_hash, name, city, country, kyc_status (pending/approved/rejected), rating_avg, roles_csv
Trips (Table 4.15)	status (pending_review/approved/rejected), reject_reason, flight_number, airline
Booking (Table 4.17)	pickup_code_hash, delivery_code_hash, seal_id, receipt_path, waybill, updated_at
Messages (Table 4.20)	booking_id (FK), listing_id (FK), read_at
Market Listing (Table 4.21)	latitude, longitude, photos_paths (JSON), condition, updated_at

Add new table data dictionaries:

- OTPEntry: id, email, otp_hash, created_at, used_at, expires_at, attempts
- OTPThrottle: id, email, ip_address, created_at
- KYCProfile: id, user_id, status, doc_type, doc_url, selfie_url, passport_url, reject_reason, created_at, updated_at
- HandoverEvent: id, booking_id, type (PICKUP_OK/DELIVERY_OK), gps_lat, gps_lng, photos_paths, seal_id, ts
- WalletEntry: id, user_id, delta, balance_after, reason, ref_id, created_at
- EscrowTx: id, booking_id, amount, currency, status (hold/released/refunded), provider, created_at
- MarketOffer: id, listing_id, from_user_id, amount, currency, message, status, ts
- MarketMeetup: id, listing_id, buyer_id, seller_id, place_text, scheduled_ts, otp_hash, confirmed_ts
- BookingUpdate: id, booking_id, old_status, new_status, changed_by, ts
- FeatureFlag: id, key (unique), value, cohort, updated_at
- AuditLog: id, actor, action, entity, before (JSON), after (JSON), request_id, ts
- Dispute: id, booking_id, filed_by, reason, status
- DisputeEvidence: id, dispute_id, uploader_id, file_url, description

Chapter 5: High-Level and Low-Level Design (Pages 31-37)

Priority: HIGH

5.2.1 Assumptions and Dependencies

Update the following:

- Remove "Payment gateways" (escrow is simulated)

- "Gemini API" -> "AI LLM providers: Google Gemini (primary), NVIDIA MiniMax, OpenAI (fallback)"
- Add: AviationStack API for flight tracking
- Add: Google OAuth 2.0 for social sign-in
- Add: SMTP server for OTP email delivery (MailHog for development)

5.3 System Architecture Diagram (Figure 5.1) - REDRAW

Redraw the architecture diagram to show:

- Client Layer: Flutter Mobile App (Riverpod 3.x, Go Router, Dio HTTP, Material Design 3)
- Backend Layer: FastAPI REST API + Jinja2 Admin + SQLite (WAL) + Alembic + APScheduler
- External Services: Gemini API, NVIDIA MiniMax, OpenAI, AviationStack, Google OAuth, SMTP
- Security: JWT, OTP (bcrypt), SlowAPI Rate Limiting, CORS, TrustedHost, Audit Logging

5.3.1 Subsystem Architecture (Figures 5.2, 5.3) - REDRAW

Admin diagram: Add KYC queue, Booking management, Escrow actions, Marketplace moderation, Audit logs, Feature flags, Wallet overview, Cookie auth.

User diagram: Add Auth (Login/Signup/Google/OTP), KYC, Trips, Bookings, Marketplace, Chat, AI Assistant, Wallet, Profile, Settings (Dark Theme).

5.5 Class Diagram (Figure 5.4) - REDRAW

Redraw with all 18+ model classes: User, Trip, RequestItem, Booking, EscrowTx, WalletEntry, HandoverEvent, KYCProfile, MarketListing, MarketOffer, MarketMeetup, Review, Message, Dispute, DisputeEvidence, BookingUpdate, FeatureFlag, AuditLog, OTPEntry, OTPThrottle.

Chapter 6: Implementation and Test Cases (Pages 38-39)

Priority: CRITICAL - MAJOR REWRITE NEEDED

This chapter is the MOST OUTDATED section. The FYP-I report described only a basic prototype with frontend screens. All backend functionality is now fully implemented.

6.1.1 Authentication - REWRITE

Rewrite to include:

- OTP-based email auth (bcrypt-hashed, single-use, 300s TTL, 3 max attempts)
- Password-based registration and login (bcrypt, min 8 chars, letter+digit)
- Google Sign-In (OAuth 2.0 ID token via Google tokeninfo endpoint)
- Forgot password flow (OTP re-issue for existing accounts)
- JWT token generation (HS256, 60-min expiry, roles in claims)
- Rate limiting (3 OTP/min, 5 logins/min, 10 Google/min)
- OTP throttling per email and IP

Reference: server/app/api/routes/auth.py

6.1.2 Book Service - REWRITE

Rewrite to include full booking state machine:

- States: REQUESTED -> HOLD_PLACED -> ACCEPTED -> PICKUP_OK -> DELIVERY_OK -> RELEASED
- Terminal states: CANCELLED, EXPIRED
- Booking creation generates OTP codes (bcrypt-hashed)
- Traveler accept/decline with ownership guards
- Pickup verification: OTP + GPS + photos + seal ID
- Delivery verification: OTP + seal match + GPS + photos
- Admin escrow release/refund
- Waybill generation (SG-XXXX-XXXXXX)

Reference: server/app/domain/bookings/service.py, server/app/api/routes/bookings.py

6.1.3 Traveling - REWRITE

Rewrite to include:

- Trip creation: airports, date, capacity, fee, flight number, airline
- Airport/airline auto-complete from bundled JSON (400+ airports, 100+ airlines)
- Admin review: pending_review -> approved/rejected
- Flight tracking via AviationStack API

Reference: server/app/api/routes/trips.py, mobile/lib/features/trips/

6.1.4 Marketplace - REWRITE

Rewrite to include:

- Listing creation: title, photos, price, category, condition, location (text + GPS)
- Search filters: keyword, category, near-me radius, price range
- Offer bargaining: SENT -> COUNTERED/ACCEPTED/DECLINED/WITHDRAWN
- Private offer threads (seller + specific buyer only)
- Meetup with OTP confirmation
- Mark sold, pause/resume

Reference: server/app/domain/marketplace/service.py

NEW SECTION: 6.1.5 Admin Dashboard

Add entirely new section:

- Jinja2 server-rendered admin panel (11 HTML templates)
- Cookie-based auth with session management
- KYC approval/rejection queue
- Booking management with timeline view
- Escrow release/refund actions

- Marketplace moderation
- Wallet overview, Audit log viewer, Feature flags, User management

Reference: server/templates/, server/app/api/routes/admin.py

NEW SECTION: 6.1.6 AI Assistant

Add entirely new section:

- Multi-provider LLM: Gemini (gemini-2.0-flash), NVIDIA MiniMax, OpenAI (gpt-4o-mini)
- Strict system prompt (ShareGo-only questions)
- Read-only database tools, no write actions
- Rate limiting: 30 req/hour per user
- Safety: refuses contraband/illegal instructions

Reference: server/app/api/routes/ai.py

NEW SECTION: 6.1.7 In-App Messaging

Add: Polling-based messaging, booking/listing context links, read receipts.

Reference: server/app/api/routes/messages.py

NEW SECTION: 6.1.8 Wallet System

Add: Simulated wallet ledger, escrow credits/debits, transaction history, running balance.

Reference: server/app/domain/escrow/service.py

NEW SECTION: 6.2 Test Cases

Add test case documentation:

- Auth: OTP request/verify, JWT claims, user creation
- Bookings: Full lifecycle (create -> accept -> pickup -> delivery -> release)
- Marketplace: Listing -> offer -> counter -> accept -> meetup -> confirm -> mark sold
- Ownership guards: 403 for unauthorized actions
- Rate limiting: 429 for exceeded limits
- Invalid transitions: 400 for incorrect state changes

Reference: server/tests/

Chapter 7: Conclusions and Future Work (Page 40)

Priority: CRITICAL - COMPLETE REWRITE

7.1 Conclusion - REWRITE

Current text says "prototype demonstrates feasibility." Replace with:

"The full ShareGo system has been implemented end-to-end. The backend includes 15+ API route modules, 18+ database tables with Alembic migrations, comprehensive security (JWT, OTP, rate limiting, audit logging), and a full admin dashboard. The mobile app covers all user flows with Material Design 3 theming (dark/light mode), 120Hz refresh rate support, and Riverpod state management. Both Feature A (personal shopping with simulated escrow) and Feature B (marketplace with bargaining) are fully functional and tested."

7.2 Future Work - REPLACE COMPLETELY

The current 4 items are ALL COMPLETED. Replace with:

- Real payment integration (Stripe/JazzCash/EasyPaisa)
- Real-time WebSocket chat (replace polling)
- Push notifications via Firebase Cloud Messaging
- Production deployment (PostgreSQL, cloud hosting)
- Advanced AI with RAG (Retrieval-Augmented Generation)
- Automated KYC verification (OCR + face matching)
- iOS App Store submission
- Analytics and monitoring dashboard
- Multi-language support (Urdu, Arabic)
- Advanced ML-based matching algorithm
- Automated booking expiry via scheduler
- End-to-end encryption for messages

Figures and Diagrams to Update

Figure	Current	Action Required
Fig 4.1: Login/Signup	Old UI	Replace with current screens (email+password+Google+OTP)
Fig 4.2: Item Listing	Old UI	Replace with current marketplace list (include dark theme)
Fig 4.3: Book Service	Old UI	Replace: date, traveler list, detail, form, review, OTP
Fig 4.4: Luggage Capacity	Old UI	Replace: flight, capacity, summary, success screens
Fig 4.5: Marketplace	Old UI	Replace: create listing, offer thread, meetup screens

Fig 4.6: ER Diagram	~8 tables	REDRAW with all 18+ tables and relationships
Fig 5.1: Architecture	Basic 3-layer	REDRAW: Flutter+Riverpod -> FastAPI+Jinja2 -> SQLite+APIs
Fig 5.4: Class Diagram	Basic	REDRAW with all 18+ model classes

New Diagrams to Add

- Booking State Machine: REQUESTED -> HOLD_PLACED -> ACCEPTED -> PICKUP_OK -> DELIVERY_OK -> RELEASED
- Marketplace Offer State Machine: SENT -> COUNTERED/ACCEPTED/DECLINED/WITHDRAWN
- Trip Status Machine: pending_review -> approved / rejected
- Authentication Flow Diagram: OTP, Password, and Google SSO flows
- Admin Dashboard Screenshots
- Dark Theme Comparison: Side-by-side light/dark screenshots

Bibliography - Add New References

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Change Priority Summary

Chapter	Priority	Effort
Ch 1: Introduction	Low	Minor text updates
Ch 2: Project Vision	Medium	Expand scope, stakeholders, objectives

Ch 3: Literature Review	None	No changes needed
Ch 4: SRS	High	New features, use cases, ER diagram, data dictionary
Ch 5: Design	High	Redraw architecture, class diagram, subsystem diagrams
Ch 6: Implementation	Critical	Complete rewrite with all implemented modules
Ch 7: Conclusions	Critical	Rewrite conclusion, replace future work